

Thomas Hsu

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Education

University at Buffalo, School of Engineering and Applied Sciences

Anticipated graduation: May 2026

Bachelor of Science in Computer Science

GPA: 3.7/4.0

Relevant Coursework: Data Structures, Algorithms & Complexity, Web Applications, Software Quality in Practice

Work Experience

Freelance Web Developer | Johnny Novo on Instagram and TikTok

July 2025 - Present

TypeScript, React, Google Firebase, Vitest

New York, NY

- Built and deployed a React-based content management system for managing and publishing ranked datasets, featuring interactive components and an admin interface powered by Google Firebase, serving ~200 daily visitors
- Reduced Firestore operation costs by 32% through implementing data denormalization, caching, and optimistic UI updates
- Implemented role-based CRUD workflows backed by Firebase, to support scalable and dynamic site content management
- Developed and utilized a CI/CD pipeline using GitHub Actions, Vitest, and Netlify to enable continuous testing, automated build validation, and rapid production deployments
- Engaged directly with the client to transform concepts into features, providing comprehensive development updates

Software Engineering Intern | Consulta Immigration

June 2025 - August 2025

TypeScript, React, ExpressJS, MongoDB, AWS, GCP, Jest

New York, NY

- Benchmarked OCR and LLM APIs (AWS Textract, OpenAI GPT-4o, etc.) for performance and cost, producing insights that led to choosing the most cost-effective option for information extraction and filtration
 - Built an automated pipeline to validate user inputs by integrating Google Gemini to extract information from client documents queried via MongoDB, cutting manual review time by 20%
 - Engineered a document filtering workflow powered by Google Gemini to identify and remove irrelevant pages from client uploads for admins with over 93% accuracy, validated using Jest unit testing
 - Architected ExpressJS API routes with a NextJS frontend to power a referral system, boosting customer acquisition efforts
 - Practiced Agile methodologies, participating in stand-ups and story planning to achieve timely delivery of 5 features
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Technical Projects

Project Management Tool | *TypeScript, NestJS, React, PostgreSQL, Docker, WebSockets*

- Developed a collaborative project management platform with real-time Kanban board functionality using NestJS and React
- Implemented JWT-based authentication, role-based access control, and WebSocket integration for live project updates

Wurdle Royale | *Python, Flask, MongoDB, Docker, WebSockets*

- Designed a containerized Flask app to build multiplayer Wordle with 30 players, developed collaboratively in a team of four
- Integrated MongoDB to manage persistent storage of user data and handle secure authentication through session tokens

TCP Python Web Server | *Python, MongoDB, Docker, WebSockets*

- Built a TCP-based HTTP/WebSocket server implementing HTTP routing, multipart parsing, and WebSocket protocol support
 - Enabled real-time messaging, authentication, large-file media uploads, and WebRTC video calls, containerized with Docker
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Technical Skills

Programming languages: JavaScript, TypeScript, Python, Java, C, C++

Frameworks & Libraries: ExpressJS, NestJS, Flask, FastAPI, React, NextJS, Tailwind, PyTest, Jest, Vitest

Developer Tools: Git, SQL, MongoDB, Docker, Amazon Web Services (AWS), Google Cloud Platform (GCP)